ECHO: Engagement & Churn Health Observatory

An End-to-End Data Pipeline and Machine Learning Analysis of Patient Retention in Digital Health

Nicolò Dalmasso

Portfolio Project — Data Science, Melbourne, Australia

github.com/ndalmasso/echo

Abstract

We present Echo, an end-to-end data engineering and machine learning pipeline for modelling patient engagement and churn risk in a digital health context. Using synthetic electronic health record (EHR) data from the Synthea generator, we implement a medallion architecture (Bronze → Silver → Gold) on Databricks using dbt, producing a validated feature table of 117 patients and 9 behavioural features. K-Means clustering identifies four clinically meaningful patient segments, and a LightGBM classifier predicts churn with AUC-ROC = 0.737. SHAP explainability analysis identifies age, income, and financial burden as the primary drivers of patient disengagement.

Keywords: dbt · Databricks · Delta Lake · patient churn · K-Means clustering · LightGBM · SHAP explainability

1 Introduction

Patient retention is a central operational challenge for digital health companies. Unlike traditional healthcare settings, telehealth and app-based platforms must actively maintain engagement: patients can disengage silently, with no formal discharge event to signal their departure. Early identification of at-risk individuals is therefore critical, both for clinical continuity and commercial sustainability. Telehealth and app-based healthcare providers (e.g., AI scribing platforms, weight management services, and digital GP services) rely on behavioural data to understand who is at risk of disengagement, what distinguishes different engagement profiles, and how resources should be allocated for targeted outreach. These are applied data science problems that require robust data infrastructure alongside interpretable predictive models. Echo addresses this by constructing a full production-style pipeline from raw EHR data to actionable outputs, implemented on the Databricks Lakehouse with dbt for transformation. The analysis delivers two outputs: a patient segmentation model identifying distinct engagement profiles, and a churn risk classifier with SHAP-based explainability. The report is structured as follows. Sec. 2 describes the data and cohort; Sec. 3 details the pipeline architecture and feature engineering; Sec. 4 presents exploratory findings; Sec. 5 covers the segmentation analysis; Sec. 6 the churn model; and Sec. 7 discusses the business implications.

2 Data

2.1 Synthea Synthetic EHR Data

All patient data is sourced from **Synthea** [Walonoski et al., 2018], an open-source synthetic EHR generator by MITRE Corporation that produces realistic, de-identified records modelled on US clinical workflows. No real patient information is used at any stage. The cohort comprises 117 patients with Massachusetts demographic profiles.

2.2 Churn Definition

A patient is classified as *churned* if they have had no recorded clinical encounter in the 365 days preceding the analysis date. This threshold aligns with annual review cycles common in primary care. Under this criterion, 20 of 117 patients (17.1%) are labelled as churned.

3 Data Pipeline

3.1 Medallion Architecture

Echo implements a *medallion architecture* on Databricks, in which data is progressively refined across three layers, each materialised as a Delta table via dbt models.

Bronze Raw Synthea CSV files ingested via PySpark.

Silver Cleaned intermediates including patient–encounter joins, inter-visit interval computation, and per-patient condition aggregations.

Gold A single feature table (`fct_patient_features`), one row per patient, engineered for direct ML consumption.

Table 1. Gold layer feature table schema.

Feature	Description
`age`	Age at analysis date (yr)
`total_encounters`	Total clinical visits
`avg_days_between_visits`	Mean inter-visit interval (d)
`condition_count`	Total diagnosed conditions
`unique_condition_count`	Distinct condition codes
`active_condition_count`	Unresolved conditions
`encounter_type_diversity`	Distinct encounter classes
`avg_claim_cost`	Mean cost per visit (USD)
`income`	Annual income (USD)
`is_churned`	Target: 1 if no visit in 365 d

3.2 Feature Engineering

Nine features are derived from the Silver layer (Tab. 1), spanning three established dimensions of patient behaviour: *recency*, *frequency*, and *clinical complexity*.

3.3 Data Quality

dbt schema tests were applied at both Bronze and Gold layers, covering primary key uniqueness on `patient_id`, non-null constraints, and accepted value ranges for encounter types. All 117 records passed with no missing values in the final feature table.

4 Exploratory Analysis

4.1 Demographics and Engagement

Fig. 1 shows that churned patients are concentrated in the 50–80 age range, while younger patients tend to remain active. This is consistent with older individuals facing greater logistical barriers to sustained care, or disengaging in the wake of a significant health event. The encounter frequency distribution (Fig. 2) reinforces visit frequency as a meaningful retention signal: active patients show higher and more varied engagement rates than churned individuals.

4.2 Clinical Complexity and Visit Regularity

Perhaps unexpectedly, higher condition counts do not appear to protect against churn (Fig. 3). The most clinically complex patient segments show the highest disengagement rates, possibly reflecting care fatigue or systemic difficulties in managing patients with multiple chronic conditions over the long term. Churned patients also tend to exhibit longer inter-visit intervals (Fig. 4), a gradual widening of visit gaps before full lapse that could serve as an early warning signal for proactive outreach.

4.3 Feature Correlations and Churn by Age

The correlation matrix (Fig. 5) reveals expected relationships: age with condition count (0.61) and encounter diversity (0.61), and total encounters with conditions (0.67). Of note, `avg_claim_cost` shows the highest individual correlation with churn (0.29), anticipating its importance in the SHAP analysis. Stratifying by age group (Fig. 6), the 65+ cohort shows the highest disengagement rate at 30.4%, while patients under 35 show the lowest, consistent with greater digital health adoption among younger adults.

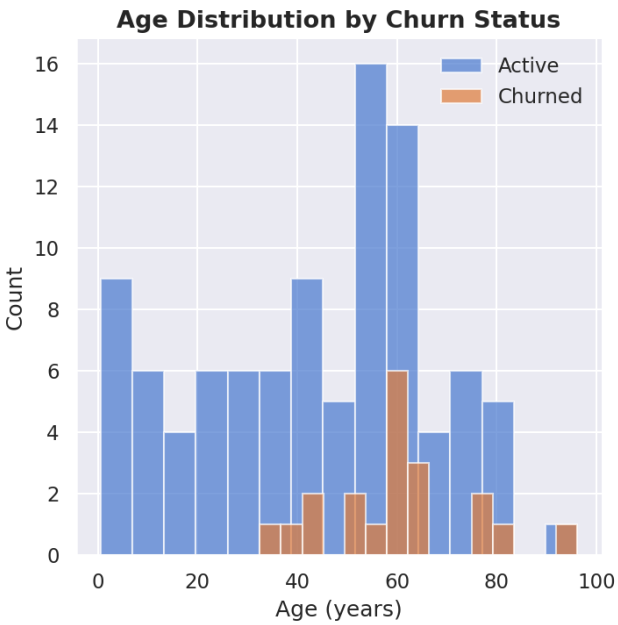

Figure 1. Age distribution by churn status.

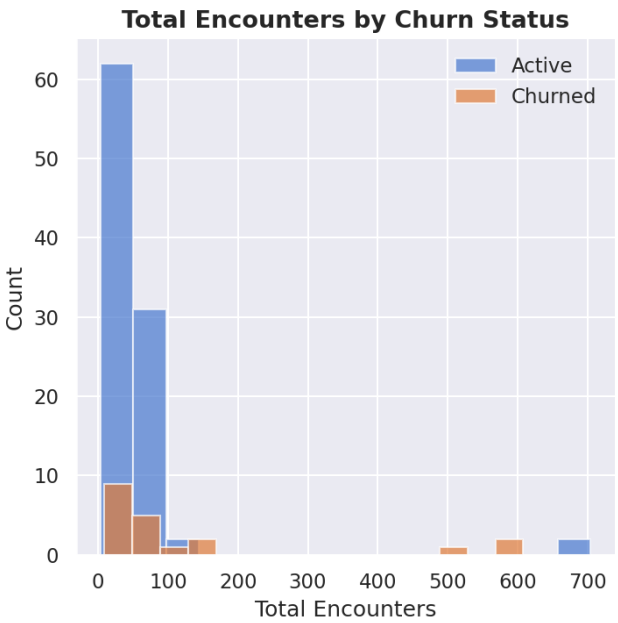

Figure 2. Total encounter distribution by churn status.

5 Patient Segmentation

5.1 Methodology

K-Means clustering is applied to six standardised behavioural features: `age`, `total_encounters`, `avg_days_between_visits`, `condition_count`, `encounter_type_diversity`, and `avg_claim_cost`. The optimal $k = 4$ is selected via the elbow method and silhouette scores across $k \in [2, 7]$. UMAP [McInnes et al., 2018] projects the feature space into two dimensions

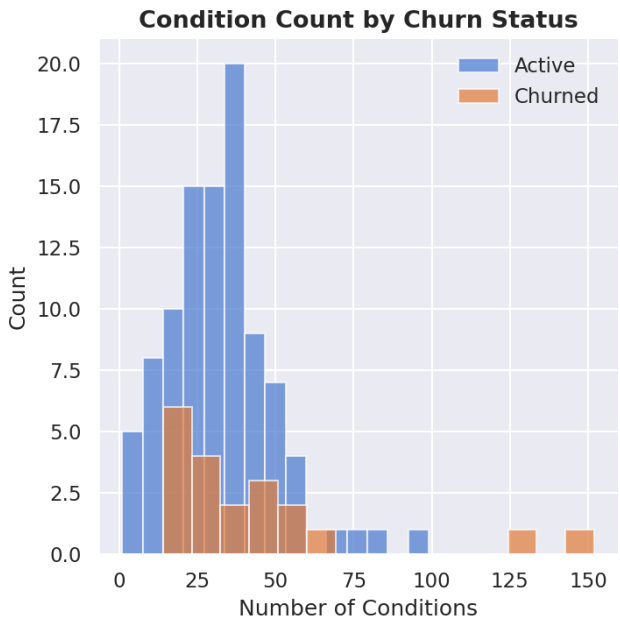

Figure 3. Condition count distribution by churn status.

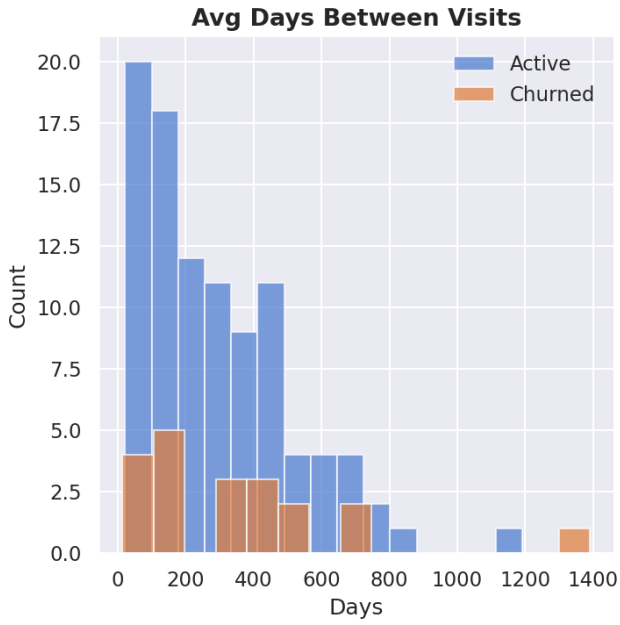

Figure 4. Average inter-visit interval by churn status.

for visualisation.

5.2 Segment Profiles

Tab. 2 reports the mean feature values and churn rates for each segment. *Stable Users* (S0) represent the core retained cohort: middle-aged patients with moderate engagement and a 19% churn rate. *Healthy Engagers* (S1) are younger with lower condition burdens and near-zero churn, making them well-suited for preventive digital interventions. *Lost High-Need* (S2) are fully disengaged despite historically high encounter volumes and substantial clinical complexity. *At-Risk Complex* (S3) are elderly, extremely high-utilisation patients with 60% churn and constitute the highest-priority outreach target. A UMAP projection of the feature space confirms these profiles visually: *Healthy Engagers* form a clearly separated group, while at-risk patients are distributed within the *Stable Users* mass, suggesting their churn risk is not apparent from individual features alone and requires the full multivariate representation to detect.

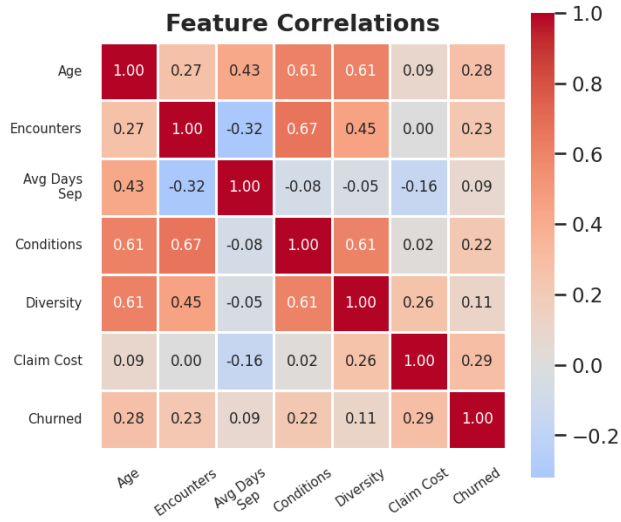

Figure 5. Feature correlation matrix across predictors and churn label.

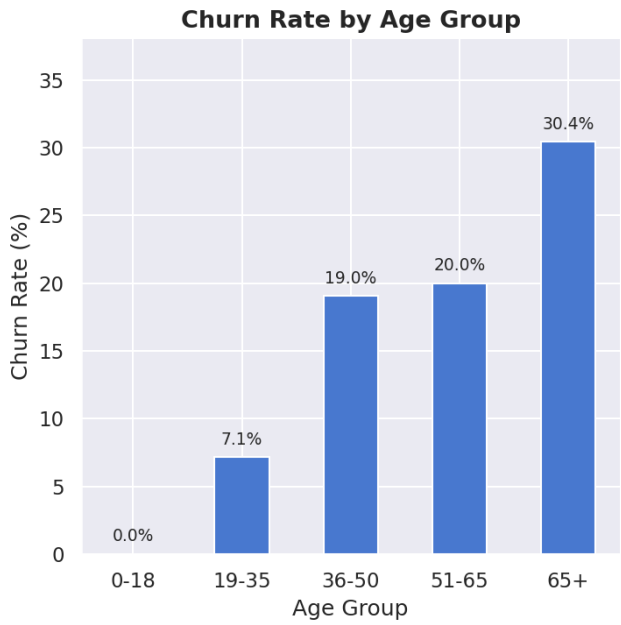

Figure 6. Churn rate by age group.

6 Churn Prediction

6.1 Model and Performance

A **LightGBM** [Ke et al., 2017] binary classifier is trained on the nine Gold features with an 80/20 stratified split and balanced class weights. Tab. 3 reports test-set performance on 24 held-out patients. An AUC-ROC of 0.737 indicates the model correctly ranks a churned patient above an active one approximately 74%

Table 2. Patient segment profiles (cluster means).

	S0	**S1**	**S2**	**S3**
Label	Stable	Healthy	Lost	At-Risk
n	70	39	3	5
Age (yr)	59.1	20.1	59.5	68.7
Encounters	49	37	118	620
Conditions	26	16	45	72
Claim Cost ($)	2,725	2,979	26,683	986
Churn Rate	19%	3%	100%	60%

Table 3. LightGBM test-set performance (24 patients).

Class	**Prec.**	**Rec.**	**F1**	n
Active	0.86	0.90	0.88	20
Churned	0.33	0.25	0.29	4
AUC-ROC			**0.737**	

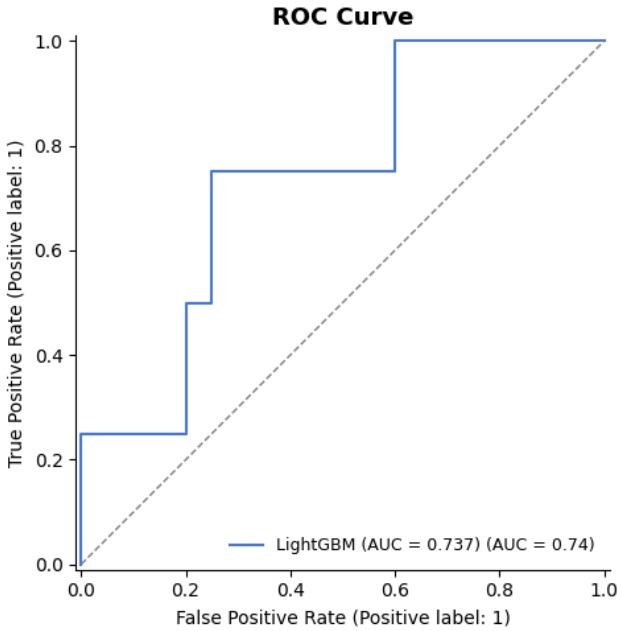

Figure 7. ROC curve on the held-out test set (AUC = 0.737). Dashed line: random classifier.

of the time. The modest recall on the churned class is expected given the small test set; both metrics would improve with a larger cohort.

6.2 Feature Importance (SHAP)

SHAP analysis [Lundberg & Lee, 2017] (Fig. 8) identifies `age` as the strongest churn predictor, followed by `income`, `total_encounters`, and `avg_claim_cost`. Age and income together dominate the model, suggesting that demographic and socioeconomic factors drive disengagement more than clinical complexity alone. The presence of `avg_claim_cost` in the top four has direct implications for subsidy programmes and proactive cost communication as retention tools.

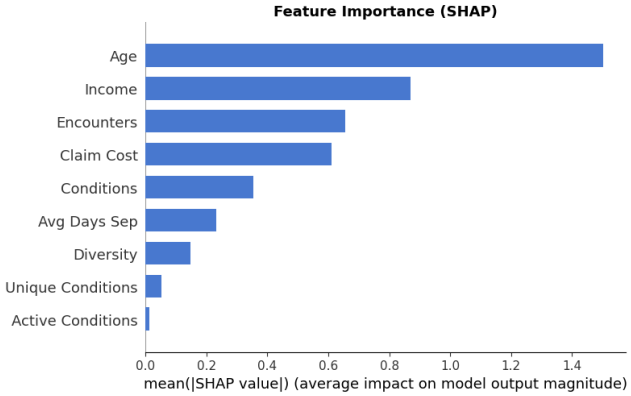

Figure 8. SHAP feature importance ranked by mean absolute contribution.

7 Business Implications

The segmentation and churn analyses together provide an actionable view of the patient population. Each segment calls for a different response: *Healthy Engagers* suit preventive nudges and digital outreach; *At-Risk Complex* patients require proactive clinical coordination; *Lost High-Need* patients, though few in number, carry the highest clinical and commercial risk and warrant direct re-engagement. The SHAP analysis highlights age and income as the dominant disengagement drivers, suggesting that demographic and socioeconomic targeting should inform outreach prioritisation. The prominence of `avg_claim_cost` in the top four predictors supports means-tested financial support and proactive cost communication as complementary retention tools. The `avg_days_between_visits` feature captures the gradual widening of visit gaps before full lapse, providing a near-real-time signal that enables outreach before disengagement becomes irreversible. The Databricks and dbt stack scales naturally from this prototype to production-scale populations, and the medallion architecture readily accommodates new data sources such as app engagement logs or telehealth session records as additional Silver layer models.

8 Conclusion

Digital health platforms face a structural challenge that traditional clinical settings do not: patients can disengage silently, with no formal signal of their departure. The central question this work addresses is whether routine EHR data (visit history, clinical complexity, cost, and demographics) contains enough signal to identify at-risk patients before they fully lapse, and whether that signal can be extracted through a production-grade pipeline suitable for operational deployment.

Using a synthetic cohort of 117 patients, we showed that the answer is yes, within the limits of the data. A medallion architecture on Databricks with dbt transforms raw records into a validated feature table from which K-Means clustering recovers four clinically interpretable engagement profiles, and a LightGBM classifier

achieves AUC-ROC = 0.737. SHAP analysis identifies age, income, and financial burden as the dominant disengagement drivers, shifting the focus of retention strategy from clinical intervention toward socioeconomic support.

The pipeline is designed to generalise. The medallion architecture accommodates new data sources as additional Silver layer models, and the methodology extends naturally to larger cohorts where model performance would improve substantially. The approach presented here could be applied directly to real patient data at telehealth providers, weight management platforms, and AI-assisted clinical services, providing an operational early-warning system for patient disengagement at scale.

Acknowledgements

Synthetic patient data provided by the Synthea project, MITRE Corporation. No real patient data was used at any stage.

Data Availability

The Synthea dataset is freely available at synthea.mitre.org/downloads. All pipeline code, dbt models, analysis notebooks, and this report are available at github.com/ndalmasso/echo.

References

Ke, G., Meng, Q., Finley, T., et al. 2017, *LightGBM: A Highly Efficient Gradient Boosting Decision Tree*, Advances in Neural Information Processing Systems, 30.

Lundberg, S. M., & Lee, S.-I. 2017, *A Unified Approach to Interpreting Model Predictions*, Advances in Neural Information Processing Systems, 30.

McInnes, L., Healy, J., & Melville, J. 2018, *UMAP: Uniform Manifold Approximation and Projection*, arXiv:1802.03426.

Walonoski, J., Kramer, M., Nichols, J., et al. 2018, *Synthea: An Approach for Generating Synthetic Patients*, Journal of the American Medical Informatics Association, 25, 230.